
Spatial Narrative Fingerprints in LLM-Generated Built-Space Stories

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Abstract

Large language models are increasingly used to generate stories, synthetic route descriptions, and planning traces for embodied agents. Yet fluent spatial prose may still encode movement, landmarks, and perspective differently from human-authored spatial language. We test whether LLM stories about built-space movement exhibit a measurable spatial-narrative grammar. We build a balanced corpus of 114 human texts and 114 real GPT-5.4-MINI generations: 80 R2R indoor route cases and 34 spatial WRITINGPROMPTS prose-control cases. We extract transparent regex/count features covering length and generic style, spatial grammar, and narrative embellishment, then evaluate feature differences and logistic-regression author classifiers under stratified 5-fold cross-validation. Human and LLM texts are highly separable. Across the full corpus, all features reach 97.4% accuracy, all non-length features reach 96.5%, and spatial-only features reach 87.7% with permutation $p = 0.0099$. The WRITINGPROMPTS control also remains separable: spatial-only features reach 83.8% accuracy and all non-length features reach 94.1%. The dominant GPT-5.4-MINI profile is first-person, sensory, formulaic, landmark-reusing, and less imperative than human route instructions. These results support a measurable LLM spatial-narrative fingerprint under the tested prompting setup, while stopping short of claims about stable cognitive maps or model-internal spatial representations.

1 Introduction

Why spatial narrative? LLMs can now write polished stories about moving through buildings, following routes, and reacting to landmarks. This makes them useful for synthetic narrative data, interactive-fiction content, and language traces for embodied or simulated agents. The risk is that fluent prose can hide systematic differences in how a model introduces places, preserves route order, marks perspective, and ends movement sequences.

Our main question. We ask whether LLM-generated stories about built-space movement show a measurable spatial-narrative grammar that differs from human spatial language. The question is narrower than general synthetic-text detection. We do not ask only whether generated text is longer, smoother, or more formulaic. We ask whether spatial and narrative features—landmarks, motion verbs, route-step density, egocentric deixis, survey terms, closure, and reuse—separate human and model-authored spatial texts.

Existing work gives us two halves of the problem. Vision-and-language navigation datasets such as R2R provide human route instructions grounded in indoor spaces Anderson et al. [2018], and later work shows that generated instructions can support navigation agents Fried et al. [2018]. Separately, human-vs-LLM style studies show that modern LLMs have measurable grammatical and rhetorical fingerprints Reinhart et al. [2025]. What is missing is a targeted corpus experiment that measures spatial-narrative structure rather than broad style alone.

Paired corpora	LLM generation	Feature grammar	Evaluation
80 R2R route cases and 34 spatial WRITINGPROMPTS prompts	GPT-5.4-MINI stories with fixed seed, temperature 0.7, and cached prompts	Length/generic, spatial, and narrative feature families	Feature tests, classifier ablations, permutation tests, and LONGSTORY transfer

Figure 1: Overview of the experiment. We compare human spatial texts with matched GPT-5.4-MINI generations, extract interpretable spatial and narrative features, and test which feature families separate author class.

Our approach. Figure 1 summarizes the study. We create a paired benchmark from two sources. In R2R, we sample 80 human indoor route instructions and ask GPT-5.4-MINI to produce a first-person micro-story that preserves the route order and stopping point. In WRITINGPROMPTS, we select 34 prompts with spatial or built-environment terms and compare human prose excerpts against prompt-matched GPT-5.4-MINI stories. We then extract interpretable feature families and evaluate both statistical differences and supervised author separability.

Quantitative preview. The main result is positive but qualified. Across all 228 texts, all features classify human vs. LLM authorship at 97.4% accuracy, all non-length features reach 96.5%, and spatial-only features reach 87.7% with permutation $p = 0.0099$. In the smaller prose-control subset, where both sides are stories rather than route instructions, spatial-only features still reach 83.8% accuracy and all non-length features reach 94.1%. The strongest LLM-leaning features are first-person language, sensory terms, formulaic phrases, and landmark reuse; human-leaning features include imperative sentence share and survey/route ratio.

Contributions.

- We propose a transparent feature profile for measuring spatial-narrative grammar in built-space narratives.
- We construct a balanced human/GPT-5.4-MINI corpus from 80 R2R route cases and 34 spatial WRITINGPROMPTS cases.
- We conduct feature-family ablations showing that spatial and narrative features separate authorship beyond length alone.
- We qualify the result with a prose-control subset and an external LONGSTORY sanity check, showing that the fingerprint is partly prompt- and genre-conditioned.

The rest of the paper reviews related work in section 2, describes the corpus and evaluation protocol in section 3, presents results in section 4, and discusses limits and implications in section 5.

2 Related Work

Human route language and vision-language navigation. R2R introduced a large benchmark of human-written indoor navigation instructions grounded in real Matterport3D environments Anderson et al. [2018]. This makes it a useful source of natural human route language: workers describe room transitions, turns, landmarks, and stopping points in task-directed prose. Speaker-Follower models later used generated instructions to improve VLN agents, showing that synthetic route language can affect downstream navigation behavior Fried et al. [2018]. ROOM-ACROSS-ROOM extends this line with multilingual, densely grounded route descriptions Ku et al. [2020]. Building on this work, we use R2R not to train a navigation agent, but to ask whether model-written spatial stories preserve a different linguistic profile from human route descriptions.

LLM spatial reasoning and spatial representation. Recent work suggests that spatial language fluency does not guarantee robust spatial reasoning. Visualization-of-Thought prompting improves LLM performance on navigation and grid-based spatial tasks, but the need for explicit state visualization highlights the difficulty of spatial reasoning from text alone Wu et al. [2024]. Tag Map shows a complementary approach: make scene context explicit as a text-based map that an LLM can query for grounded planning Zhang et al. [2024]. Other work compares LLM spatial preposition behavior against human spatial-representation data and finds both human-like vertical preferences and non-human axis-specific patterns Wu and Deng [2025]. Hybrid Mind similarly frames spatial cognition

Source	Human n	LLM n
R2R	80	80
WRITINGPROMPTS	34	34
Total	114	114

Table 1: Balanced corpus used for the main human-vs-LLM analysis. Each human text has a matched GPT-5.4-MINI generation from the same route or prompt source.

in terms of landmark, route, and survey knowledge, finding weak unaided LLM performance and stronger tool-augmented performance Yang et al. [2025]. Our study differs by measuring the textual structure of spatial stories, not task-answer correctness.

Human-vs-LLM style and story generation. Human-vs-LLM writing studies show that generated text can be separated from human text using grammatical and rhetorical features. Reinhart et al. find systematic differences across Biber-style features and show that instruction-tuned models diverge more from human style than base models Reinhart et al. [2025]. Story generation work provides human prose resources and evaluation concerns relevant to our control corpus. WRITINGPROMPTS pairs prompts with large-scale human-written stories Fan et al. [2018], while StoryAlign argues that story evaluation should track human preferences over coherence, creativity, characterization, fluency, and relevance Xia et al. [2026]. A recent narrative-theory survey also argues against a single generic narrative-quality benchmark and favors theory-specific attributes Liu et al. [2026]. We follow that recommendation by focusing on a narrow attribute family: spatial movement through built environments.

3 Methodology

Problem formulation. We treat each text x_i as an observation with author label $y_i \in \{\text{human}, \text{LLM}\}$ and source label $s_i \in \{\text{R2R}, \text{WRITINGPROMPTS}\}$. The goal is not to build a production detector. Instead, we ask whether transparent feature families contain enough signal to separate author class and whether the separating signal is spatial and narrative rather than only length.

3.1 Corpora and Generation

R2R route cases. We sampled 80 R2R routes with seed 42 and selected one human instruction per route. For each route, GPT-5.4-MINI generated one 160–220 word first-person micro-story that preserved the route constraints, order, and stopping point. This setting directly tests whether a model narrativizes a concrete indoor route differently from human route language, but it also introduces a known genre shift: human texts are instructions, while model texts are stories.

WRITINGPROMPTS prose control. To reduce the route-instruction vs. fiction confound, we also used 34 WRITINGPROMPTS prompt/story pairs whose prompts contained spatial or built-environment terms. Human stories were truncated to 220 words. GPT-5.4-MINI received the original prompt and was asked for a 160–220 word first-person story scene centered on embodied movement through an indoor or built space when compatible.

External sanity check. We used 80 spatial LONGSTORY excerpts as an external AI-story check. These excerpts were not used for training or cross-validation. A classifier trained on the main corpus scored them only after all main analyses were complete.

Model and reproducibility details. Generation used the OpenAI Responses API with GPT-5.4-MINI, temperature 0.7, and max output tokens 320. The run made 114 API calls, used 20,316 input tokens and 29,078 output tokens, and completed in 6 minutes 33 seconds. Based on the official GPT-5.4-MINI model and pricing pages, the input/output prices at the time of the run were USD 0.75 and USD 4.50 per one million tokens, respectively OpenAI [2026a,b]. The estimated generation cost was about USD 0.15. All prompt records are saved in `results/prompts/llm_prompts.jsonl`.

Scope	Feature family	Accuracy	F1	ROC-AUC	95% acc. CI
All	Length-only	87.7	89.0	90.9	83.3–92.5
All	Spatial-only	87.7	88.4	95.6	85.9–89.1
All	Narrative-only	94.7	94.8	97.8	92.6–96.5
All	All without length	96.5	96.6	98.3	94.7–98.7
All	All features	97.4	97.4	99.7	95.1–99.1
R2R	Spatial-only	98.1	98.2	100.0	95.6–100.0
R2R	All without length	100.0	100.0	100.0	100.0–100.0
WRITINGPROMPTS	Length-only	69.1	69.6	82.0	60.9–75.3
WRITINGPROMPTS	Spatial-only	83.8	84.1	89.1	72.9–95.7
WRITINGPROMPTS	All without length	94.1	94.3	95.8	92.5–97.1
WRITINGPROMPTS	All features	95.6	95.8	97.8	92.6–98.6

Table 2: Author-classification results under stratified 5-fold cross-validation. Values are percentages. Best result within each scope is in **bold**.

3.2 Feature Extraction

We use interpretable regex/count features implemented in `src/features.py`. We normalize many lexical counts per 1,000 tokens so that length does not dominate every comparison. The feature set has three families.

Length and generic style. These features include word count, sentence count, mean sentence length, lexical diversity, hapax ratio, nominalization-like suffixes, `-ing` participials, coordination, `that`, and passive-like forms. They provide a generic synthetic-text baseline.

Spatial grammar. These features count direction terms, cardinal/survey terms, egocentric terms, landmarks, motion verbs, sequence markers, deixis, route-step density, landmark reuse, left/right imbalance, explicit closure, and imperative sentence share. This family is the main test of the spatial-narrative grammar hypothesis.

Narrative embellishment. These features count first-person and second-person language, sensory terms, affect terms, formulaic phrases, dialogue marks, and narrative-to-route ratio. We keep this family separate because it captures the story-like packaging that may accompany spatial movement.

3.3 Evaluation

Feature tests. For planned feature comparisons, we use Mann-Whitney U tests as the primary nonparametric test, Benjamini-Hochberg correction across feature comparisons, and Cohen’s *d* plus Cliff’s delta as effect sizes. We report the most interpretable corrected differences in section 4.2.

Classifier ablations. We train standardized logistic-regression classifiers with stratified 5-fold cross-validation. We report accuracy, F1, ROC-AUC, and bootstrap 95% confidence intervals for cross-validated accuracy. The ablations are chance, length-only, spatial-only, narrative-only, all non-length features, and all features. We also run 100-label permutation tests for the aggregate spatial-only, all-without-length, and all-feature classifiers; this gives a minimum attainable *p*-value of 0.0099.

Baselines and interpretation. Chance is 50% because the corpus is balanced by author class. Length-only is an important negative-control baseline: if it performs as well as richer families, then the result is mostly a length artifact. Source-specific analyses for R2R and WRITINGPROMPTS expose genre sensitivity.

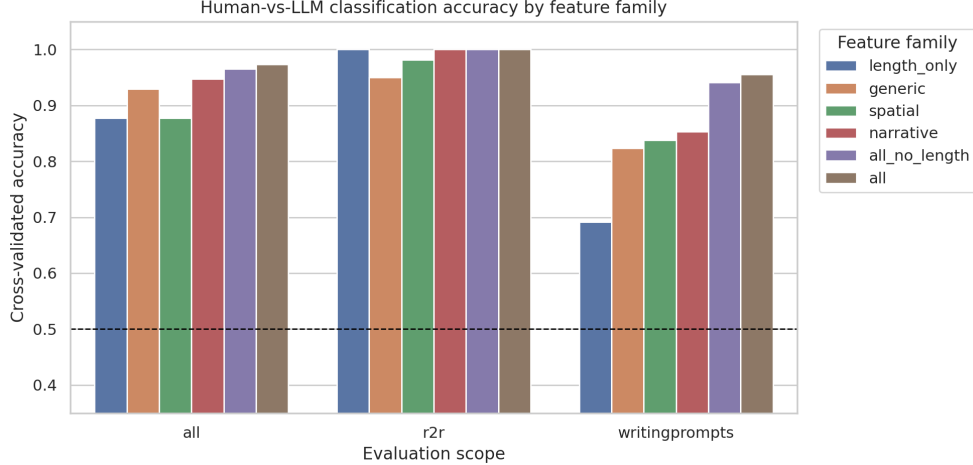


Figure 2: Classification accuracy by feature family. Spatial-only and narrative-only features both exceed the chance baseline, while all non-length features nearly match the full feature set.

Scope	Feature	Human mean	LLM mean	Direction	BH q
All	sensory_per_1000	1.77	14.67	LLM higher	5.41×10^{-32}
All	formulaic_phrase_per_1000	0.08	6.09	LLM higher	5.46×10^{-30}
All	survey_route_ratio	0.252	0.065	LLM lower	5.09×10^{-28}
All	imperative_sentence_share	0.543	0.000	LLM lower	1.50×10^{-24}
All	first_person_per_1000	20.53	74.21	LLM higher	1.62×10^{-20}
All	landmark_reuse_ratio	0.121	0.317	LLM higher	1.03×10^{-14}

Table 3: Key full-corpus feature differences after Benjamini-Hochberg correction. The largest effects show GPT-5.4-MINI using more sensory, formulaic, first-person, and landmark-reusing language, while human texts use more imperative route grammar.

4 Results

4.1 Human and LLM Spatial Texts Are Highly Separable

Figure 2 and table 2 show the main classifier results. Across the full corpus, all features reach 97.4% accuracy and 99.7 ROC-AUC. Removing length barely changes the result: all non-length features still reach 96.5% accuracy and 98.3 ROC-AUC. Spatial-only features reach 87.7% accuracy, and the permutation test gives $p = 0.0099$ for the spatial-only, all-without-length, and all-feature aggregate classifiers.

The source-specific results clarify the role of genre. In R2R, all non-length features separate the two author classes perfectly, and spatial-only features reach 98.1% accuracy. This is expected because human R2R texts are route instructions while GPT-5.4-MINI outputs are first-person stories. The more important check is WRITINGPROMPTS: length-only accuracy falls to 69.1%, but spatial-only features still reach 83.8%, all non-length features reach 94.1%, and all features reach 95.6%. This weakens the simplest explanation that the full-corpus result is only a route-instruction artifact.

4.2 The LLM Profile Is Sensory, First-Person, and Landmark-Reusing

Table 3 lists the largest and most interpretable corrected feature differences. Across the full corpus, GPT-5.4-MINI uses far more sensory language than humans (14.67 vs. 1.77 terms per 1,000 tokens, BH $q = 5.41 \times 10^{-32}$), more formulaic phrases (6.09 vs. 0.08 per 1,000, $q = 5.46 \times 10^{-30}$), more first-person language (74.21 vs. 20.53 per 1,000, $q = 1.62 \times 10^{-20}$), and more landmark reuse (0.317 vs. 0.121, $q = 1.03 \times 10^{-14}$). Humans show higher imperative sentence share and higher survey/route ratio.

Key spatial-narrative features

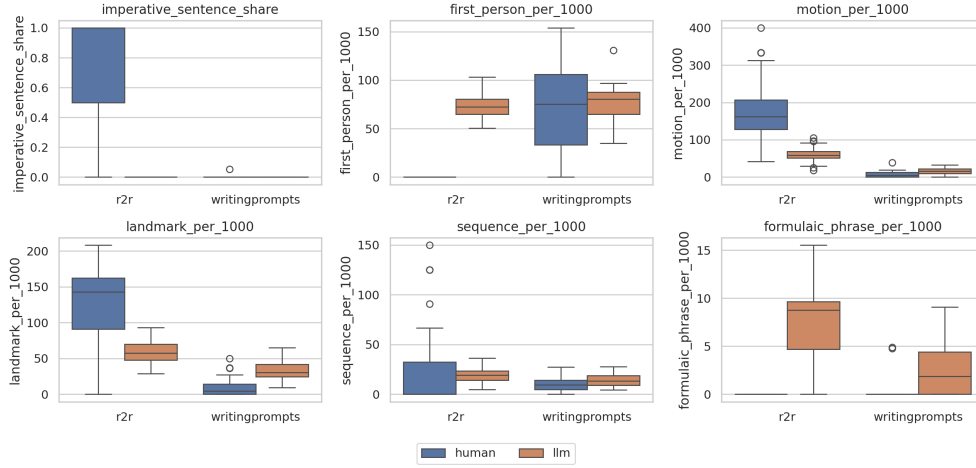


Figure 3: Distributions for selected spatial and narrative features. GPT-5.4-MINI outputs concentrate on sensory and formulaic scene language, while human R2R instructions contain more task-directed imperative structure.

Feature	Human mean	LLM mean	Direction	BH q
landmark_per_1000	9.79	32.92	LLM higher	4.62×10^{-8}
sensory_per_1000	5.93	16.75	LLM higher	1.19×10^{-7}
route_step_density	43.55	64.90	LLM higher	8.33×10^{-6}
motion_per_1000	7.81	16.92	LLM higher	7.06×10^{-5}
formulaic_phrase_per_1000	0.28	2.79	LLM higher	1.96×10^{-4}
survey_route_ratio	0.289	0.117	LLM lower	5.60×10^{-4}

Table 4: Key feature differences within the WRITINGPROMPTS prose-control subset. Both human and LLM texts are prose, yet GPT-5.4-MINI still uses more landmarks, movement language, sensory detail, and formulaic phrasing.

The WRITINGPROMPTS feature table is the strongest evidence against a pure route-instruction explanation. In this prose-control subset, GPT-5.4-MINI uses more landmarks (32.92 vs. 9.79 per 1,000, $q = 4.62 \times 10^{-8}$), more sensory language (16.75 vs. 5.93 per 1,000, $q = 1.19 \times 10^{-7}$), more route-step language (64.90 vs. 43.55, $q = 8.33 \times 10^{-6}$), and more motion verbs (16.92 vs. 7.81 per 1,000, $q = 7.06 \times 10^{-5}$). The model also has a lower survey/route ratio, suggesting a preference for embodied path progression over map-like description.

4.3 Feature Importance and External Transfer

Figure 4 and table 5 show standardized logistic coefficients for predicting LLM authorship. The top positive coefficients are first-person terms, word count, hapax ratio, formulaic phrases, sensory terms, and landmark reuse. The strongest negative coefficients include survey/route ratio, dialogue mark density, sentence count, and narrative/route ratio. The model is therefore not separating author class by a single feature; it combines perspective, style, route structure, and repetition.

The LONGSTORY sanity check shows limited transfer. A classifier trained on the main corpus predicts 43.8% of the 80 LONGSTORY excerpts as LLM, with mean LLM probability 0.412. This weak external transfer suggests that the measured fingerprint is not simply “any AI story.” It is tied to the tested model, prompts, corpus mixture, and spatial-navigation setup.

Feature	Coefficient
first_person_per_1000	1.463
word_count	1.434
hapax_ratio	1.217
formulaic_phrase_per_1000	1.089
sensory_per_1000	0.991
landmark_reuse_ratio	0.972
survey_route_ratio	-0.896
dialogue_mark_density	-0.860
sentence_count	-0.845
narrative_route_ratio	-0.835

Table 5: Top standardized logistic coefficients for predicting LLM authorship. Positive coefficients are LLM-leaning and negative coefficients are human-leaning.

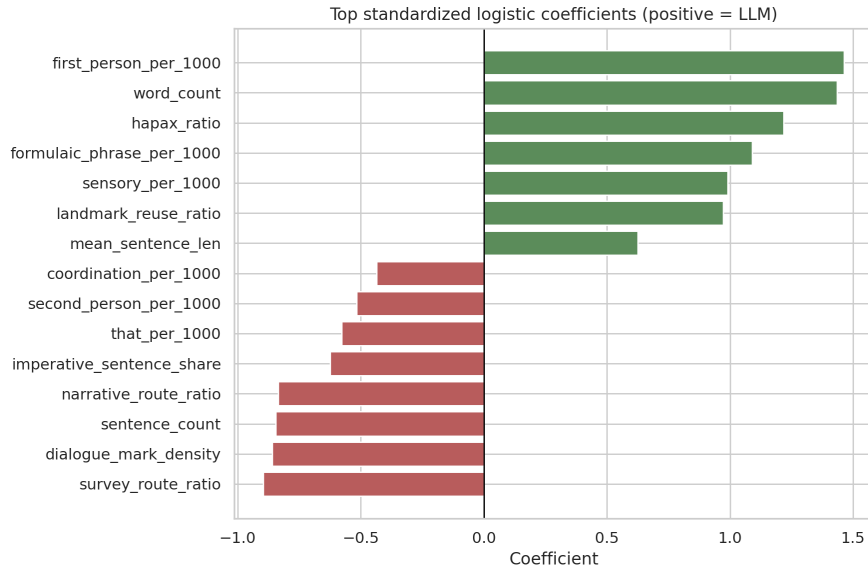


Figure 4: Top standardized logistic-regression coefficients for predicting LLM authorship. Positive coefficients are LLM-leaning; negative coefficients are human-leaning.

5 Discussion

What does the fingerprint mean? The results support a measurable LLM spatial-narrative profile under the tested conditions. The clearest pattern is that GPT-5.4-MINI turns movement through built space into an embodied scene: it writes in first person, adds sensory detail, repeats landmarks, uses formulaic scene beats, and closes routes explicitly. Human R2R language is more terse, imperative, and task-directed. The WRITINGPROMPTS control shows that this pattern does not disappear when both sides are prose, although the evidence there is based on only 34 prompt-matched pairs.

Why length is not enough. Length matters: the length-only classifier reaches 87.7% accuracy in the full corpus. But it does not explain the result. All non-length features reach 96.5% accuracy, spatial-only features reach 87.7%, and the WRITINGPROMPTS subset remains separable even when length-only performance drops to 69.1%. The feature tests also point to interpretable spatial and narrative differences, not only longer output.

What we cannot claim. This study measures textual behavior, not model cognition. The results do not prove that LLMs lack spatial maps, possess spatial maps, or use a stable internal representation of built space. They show that, under this prompt design, a current instruction-following model

produces spatial stories with a distinctive feature profile. The profile may reflect training data, RLHF/RLAIF style, prompt wording, instruction-following behavior, or genre conventions.

Limitations. The largest limitation is the R2R genre confound: human texts are route instructions, while model texts are requested stories. Spatial-only and non-length ablations reduce this concern but do not remove it. The WRITINGPROMPTS control is closer to the target question but small. The study tests one model, GPT-5.4-MINI, at one temperature and one prompt style. First-person effects partly follow from the prompt. Feature extraction is regex-based and approximate, favoring interpretability over parser-level precision. Finally, no human annotators judged route validity, spatial coherence, or narrative quality.

Implications. For synthetic data, the findings argue for measuring spatial-narrative structure before mixing model-written stories with human spatial language. For embodied-agent evaluation, they suggest that route-like prose can look coherent while encoding different landmark and perspective habits. For story-generation research, they show that domain-specific narrative attributes can reveal differences that generic quality scores or broad style classifiers may miss.

6 Conclusion

We tested whether LLM-generated built-space stories differ from human spatial language in measurable ways. Using a balanced corpus of 114 human texts and 114 GPT-5.4-MINI generations, we found strong author separability with interpretable features: 97.4% accuracy with all features, 96.5% without length, and 87.7% with spatial-only features. In a smaller WRITINGPROMPTS prose-control subset, spatial-only features still reached 83.8% accuracy and all non-length features reached 94.1%.

The main takeaway is cautious: GPT-5.4-MINI has a measurable spatial-narrative fingerprint in this setup, but the fingerprint should be read as a prompt- and genre-conditioned feature profile rather than evidence about model-internal spatial maps. Future work should collect human-written 160–220 word first-person building-navigation stories from the same prompts used for LLMs, add human judgments of spatial coherence and route validity, and compare base, instruction-tuned, and preference-tuned models across multiple model families.

References

- Peter Anderson, Qi Wu, Damien Teney, Jake Bruce, Mark Johnson, Niko Sunderhauf, Ian Reid, Stephen Gould, and Anton van den Hengel. Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018. URL <https://arxiv.org/abs/1711.07280>.
- Angela Fan, Mike Lewis, and Yann Dauphin. Hierarchical neural story generation. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, 2018. URL <https://arxiv.org/abs/1805.04833>.
- Daniel Fried, Ronghang Hu, Volkan Cirik, Anna Rohrbach, Jacob Andreas, Louis-Philippe Morency, Taylor Berg-Kirkpatrick, Kate Saenko, Dan Klein, and Trevor Darrell. Speaker-follower models for vision-and-language navigation. In *Advances in Neural Information Processing Systems*, 2018. URL <https://arxiv.org/abs/1806.02724>.
- Alexander Ku, Peter Anderson, Roma Patel, Eugene Ie, and Jason Baldridge. Room-across-room: Multilingual vision-and-language navigation with dense spatiotemporal grounding. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020. URL <https://arxiv.org/abs/2010.07954>.
- David Y. Liu, Aditya Joshi, and Paul Dawson. Narrative theory-driven LLM methods for automatic story generation and understanding: A survey, 2026. URL <https://arxiv.org/abs/2602.15851>.
- OpenAI. GPT-5.4 mini model documentation. <https://developers.openai.com/api/docs/models/gpt-5.4-mini>, 2026a. Accessed 2026-06-08.

- OpenAI. OpenAI API pricing. <https://openai.com/api/pricing/>, 2026b. Accessed 2026-06-08.
- Alex Reinhart, Ben Markey, Michael Laudenbach, Kachata Pantusen, Ronald Yurko, Gordon Weinberg, and David West Brown. Do LLMs write like humans? variation in grammatical and rhetorical styles. *Proceedings of the National Academy of Sciences*, 122:e2422455122, 2025. doi: 10.1073/pnas.2422455122. URL <https://arxiv.org/abs/2410.16107>.
- Wenshan Wu, Shaoguang Mao, Yadong Zhang, Yan Xia, Li Dong, Lei Cui, and Furu Wei. Mind’s eye of LLMs: Visualization-of-thought elicits spatial reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2024. URL <https://arxiv.org/abs/2404.03622>.
- Wenya Wu and Weihong Deng. Spatial representation of large language models in 2d scene. In *Proceedings of the Fourth Workshop on Generation, Evaluation and Metrics*, pages 18–29. Association for Computational Linguistics, 2025. URL <https://aclanthology.org/2025.gem-1.3/>.
- Haotian Xia, Hao Peng, Yunjia Qi, Xiaozhi Wang, Bin Xu, Lei Hou, and Juanzi Li. Storyalign: Evaluating and training reward models for story generation, 2026. URL <https://arxiv.org/abs/2605.04831>.
- A. Yang, Cheng Fu, Q. Jia, W. Dong, M. Ma, H. Chen, et al. Evaluating and enhancing spatial cognition abilities of large language models. *International Journal of Geographical Information Science*, pages 1–36, 2025. doi: 10.1080/13658816.2025.2490701. URL <https://doi.org/10.1080/13658816.2025.2490701>.
- Mike Zhang, Kaixian Qu, Vaishakh Patil, Cesar Cadena, and Marco Hutter. Tag map: A text-based map for spatial reasoning and navigation with large language models, 2024. URL <https://arxiv.org/abs/2409.15451>.